



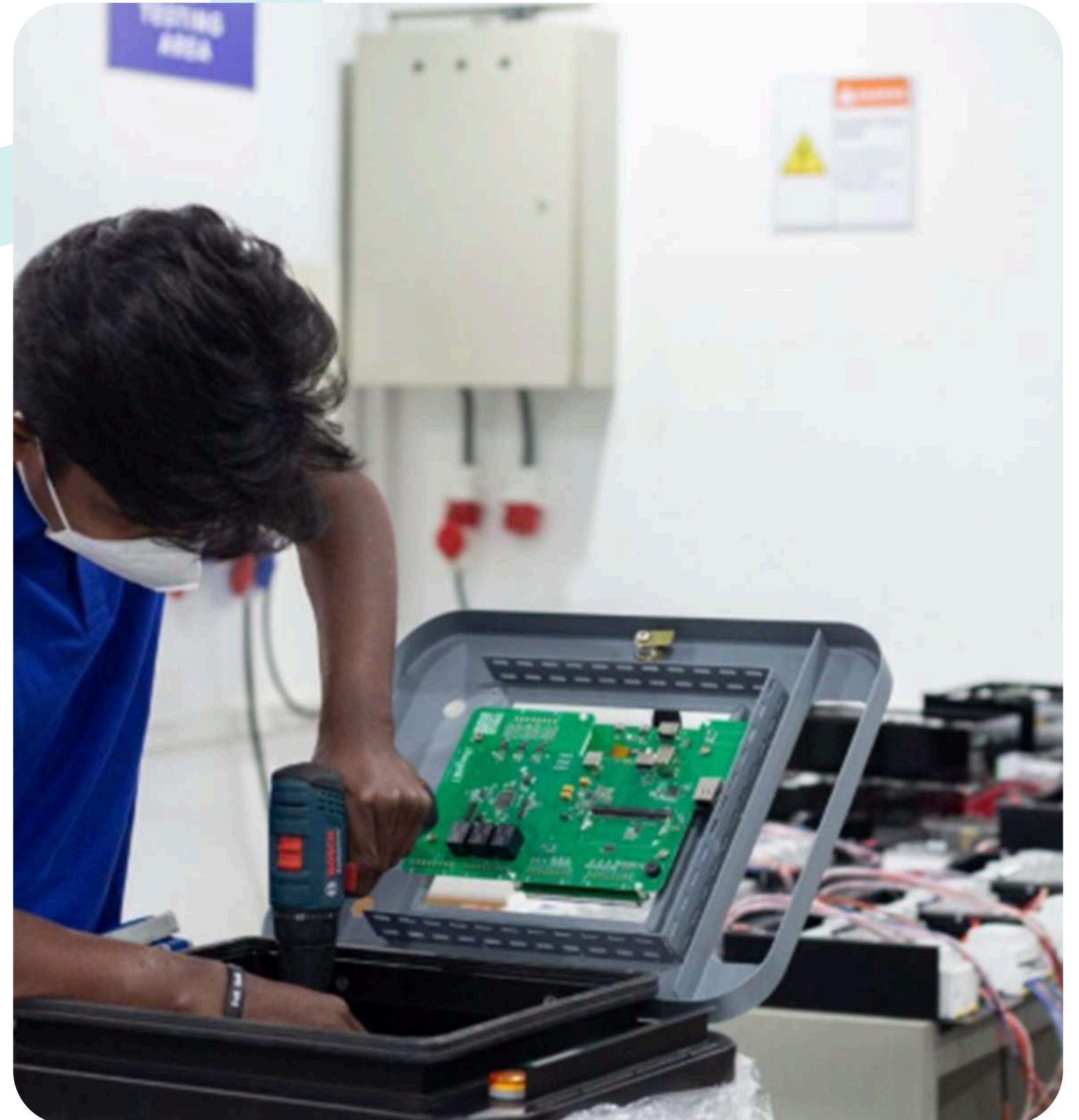
VEGA INNOVATIONS

Final Project Presentation

ME4393 – ADVANCED MANUFACTURING ENGINEERING
GROUP 08

Group Members

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Introduction

- An industrial field visit was conducted at Vega Innovations to study its product development and manufacturing environment.
- Several practical manufacturing and process-related issues were observed in these areas.



Objective

- To identify issues and develop suitable solutions using advanced manufacturing methods.



Problem 1 - Inaccurate Bending in EV Charger Chassis Manufacturing

Inaccurate bending in EV charger chassis leads to poor fit, misalignment, and increased rework during assembly.



Problem Description

The EV charger chassis is manufactured using sheet metal components that are first laser cut and then bent using manual or semi-automatic bending processes.

Current methods:

- manual bending
- press brake bending

Effects on Production

- Inaccurate bending leads to poor fit and misalignment
- High dependency on operator skill
- Multiple components require welding and assembly
- Increased rework before welding
- Longer production time and reduced consistency



Limitations of Conventional Bending & Welding

Bending (Manual / Press Brake) Limitations

- Localized force causes uneven deformation
- High springback leads to dimensional inaccuracies
- Strong dependency on operator skill
- Difficult to maintain repeatability for complex shapes

Welding Limitations

- Multiple components require welding and alignment
- Misalignment due to bending errors
- Increased rework and adjustment before welding
- Longer production time and higher labor effort



Why Hydroforming (Flexform)?

Hydroforming is an advanced sheet metal forming process that uses high-pressure fluid and a flexible diaphragm to produce accurate and complex shapes.

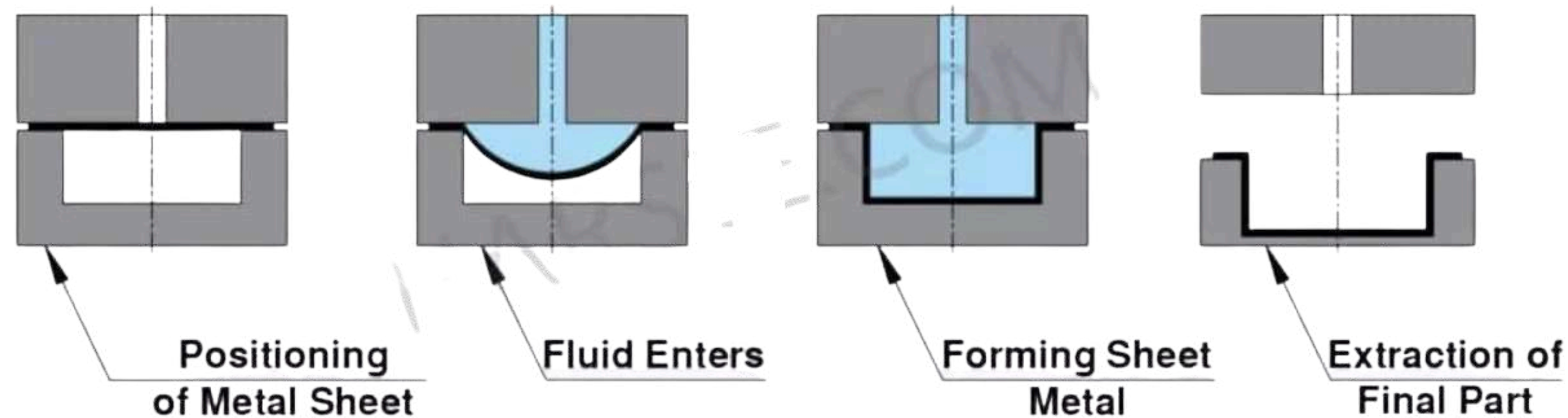
Main Advantage :

Enables integration of multiple components into a single part, reducing the need for welding and assembly.

Feature	Benefit
Uniform pressure forming	High dimensional accuracy
Biaxial stretching	Reduced stress concentration
Reduced springback	Better dimensional control
Less operator dependency	Improved consistency
Single-step forming	Reduced production time



Hydroforming (Flexform)



Working Principle

- The sheet metal blank is placed on a rigid die (4140 Alloy Steel) with the required shape
- A rubber diaphragm (Urethane) seals the chamber and holds the sheet in position
- High-pressure fluid is applied to generate uniform forming force
- The sheet is pressed into the die and deforms plastically
- The final component is formed with improved accuracy and consistency

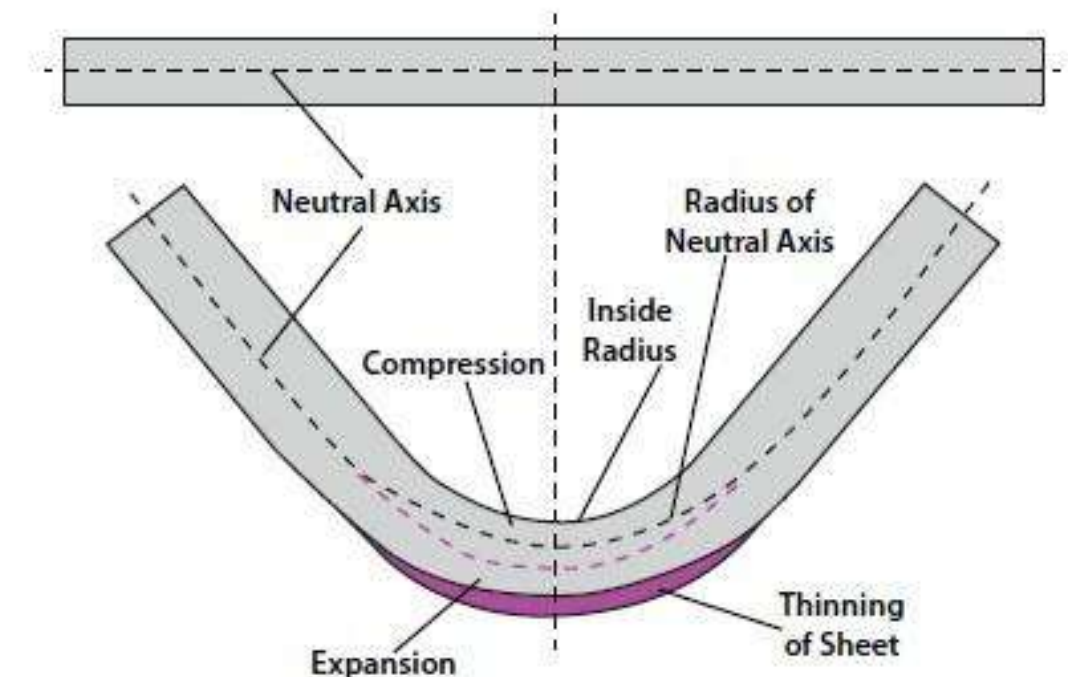
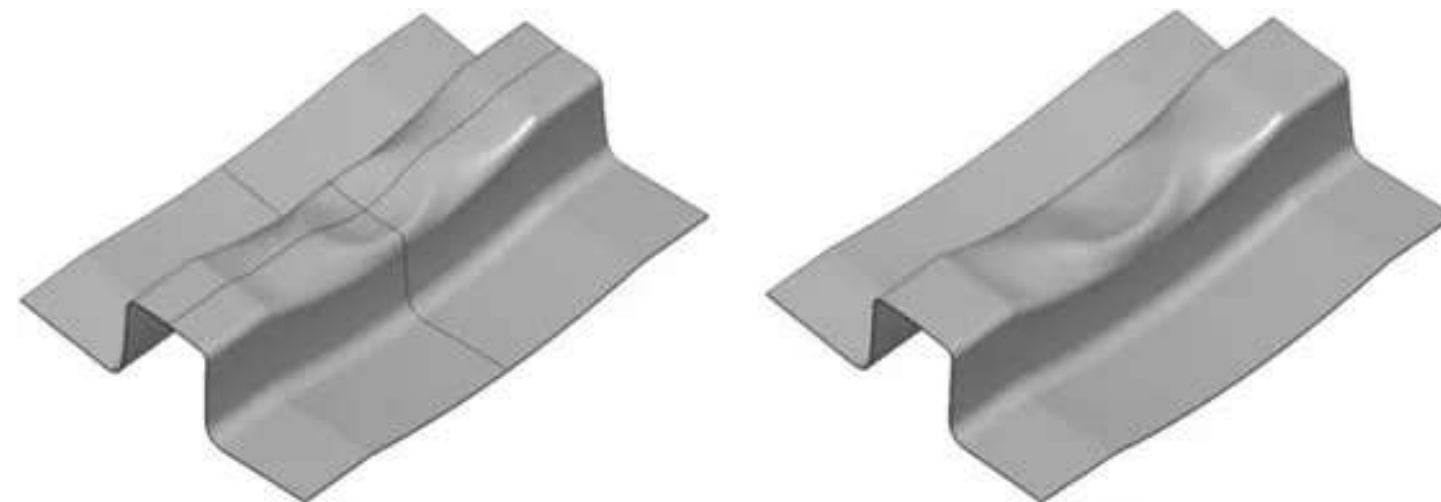
Important Phenomena & Springback

Important Phenomena

- **Thinning** → occurs in high curvature areas due to stretching of material
- **Wrinkling** → caused by compressive instability in low-tension regions
- **Tearing** → occurs when the material exceeds its forming limit

Why Hydroforming Reduces Springback

- Uniform pressure results in more even stress distribution
- Increased plastic deformation reduces elastic recovery
- Biaxial stretching improves dimensional stability





Parameters, Advantages & Limitations

Key Process Parameters

- Pressure → controls forming depth and shape accuracy
- Die design → determines final geometry of the component
- Material properties → affect formability and deformation behavior
- Lubrication → improves material flow and reduces friction
- Blank size → influences wrinkling and tearing

Advantages

- High dimensional accuracy and repeatability
- Better surface finish with reduced defects
- Reduced springback compared to bending
- Enables part integration and reduces welding
- Lower rework and improved production efficiency

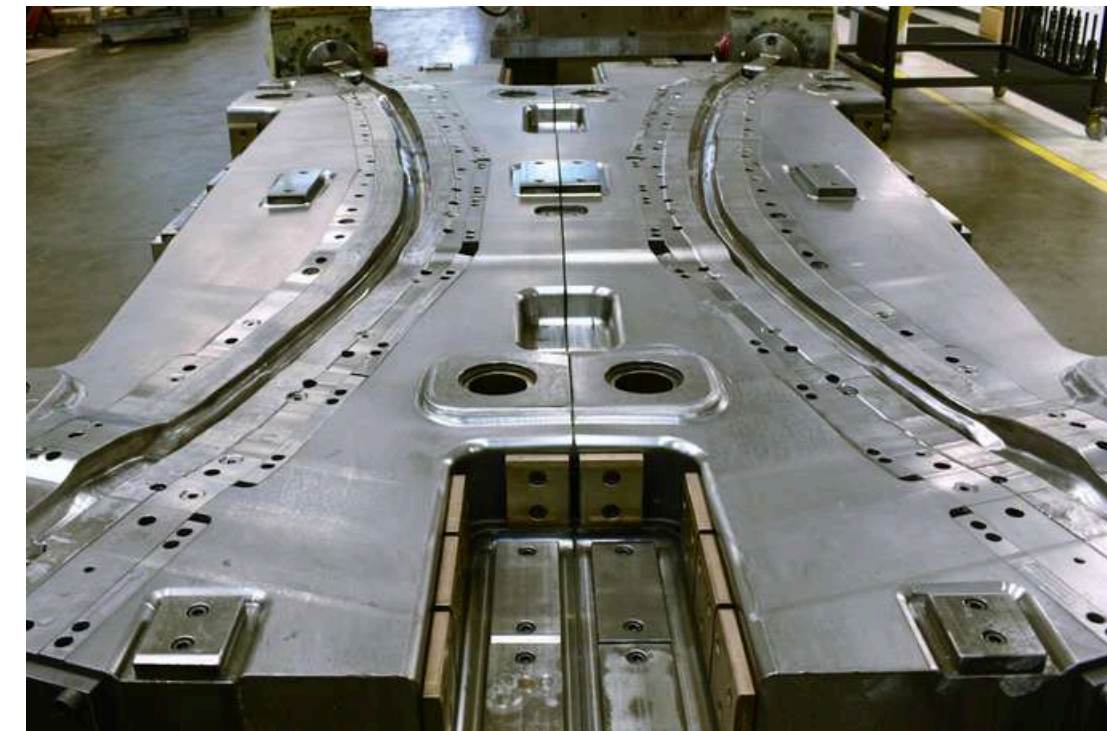


Parameters, Advantages & Limitations



Limitations

- High initial machine and tooling cost
- Requires precise die design and process control
- Not suitable for simple bending operations
- Slower compared to conventional methods for basic parts



Hydroforming vs Alternative Methods

Method	Advantages	Limitations
Press Brake Bending	Low cost, simple setup	Low accuracy, high springback, multiple operations
Incremental Sheet Forming (ISF)	Flexible, no dedicated die	Very slow, not suitable for mass production
Electromagnetic Forming (EMF)	High precision, low springback	Limited to specific materials, high cost
Deep Drawing	Good for simple shapes	Limited geometry, not suitable for complex multi-



Problem 2 - ABS (Acrylonitrile Butadiene Styrene) Sheet Cutting Issue

Poor Trimming Quality in Thermoformed ABS Parts



Problem Description

- ABS parts are used to EV charger housings and enclosures
- After thermoforming, parts require precise trimming to final geometry

Current methods:

- CNC routing
- Manual cutting

Effects on Production

- Edge melting and localised deformation near cut lines
- Poor surface finish at trimmed edges
- Sealing faces become uneven
- High rework rate increased production cost and cycle time



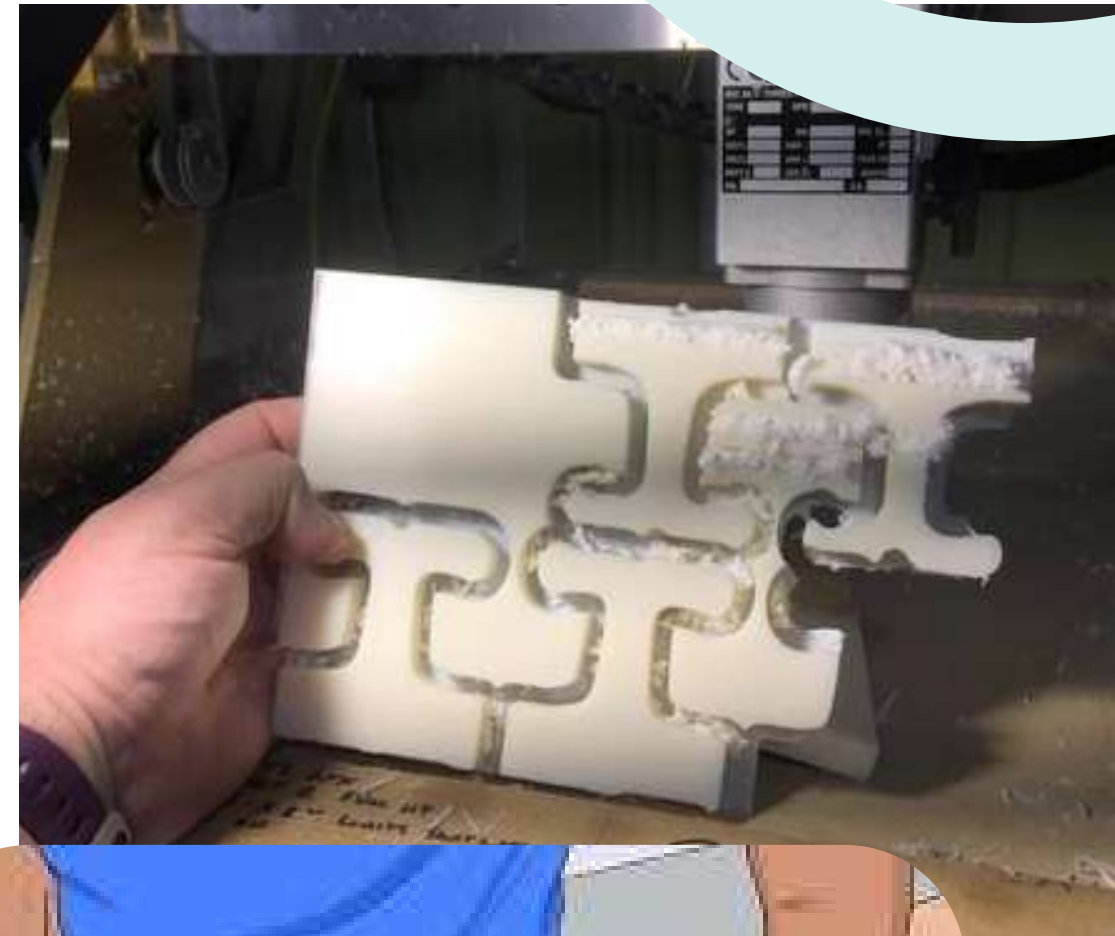
Current Method Limitations

CNC Routing:

- The cutting tool generates friction based heat during operation
- Heat builds up at the ABS surface, causing melting and burring
- chip re-welding occurs on the edges, which degrades surface quality
- Difficult to maintain tight tolerances on curved thermoformed profiles

Manual Cutting:

- Operator dependent so high dimensional variation between parts
- No proper process control, the cutting paths become inconsistent
- Not scalable for batch production



Introduction to Waterjet Cutting *

- A cold-cutting process that uses a high-pressure, high-velocity water jet to cut through materials
- No heat generated at the cut zone — no thermal damage
- Controlled by CNC for precise, repeatable cuts

Two Types:

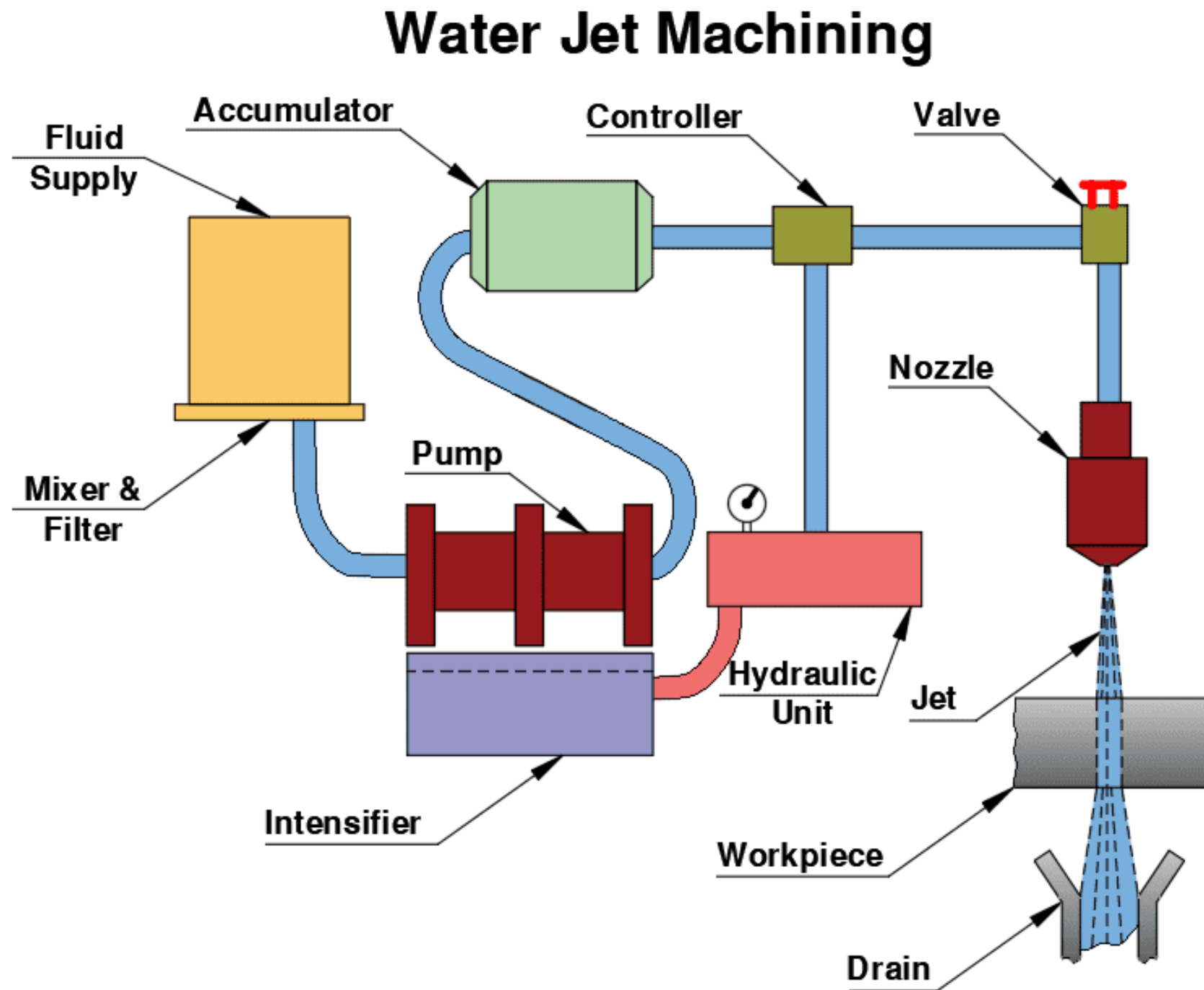
- Pure waterjet → soft materials (ABS, rubber, foam)
- Abrasive waterjet → hard materials (metal, glass)

Key Principle:

- Cutting energy comes from kinetic energy of water, not thermal energy
- No heat-affected zone (HAZ) – material properties at edge remain unchanged



System Components



- **Fluid supply & filter** – supplies clean water into the system
- **Pump** – feeds water into the system (does not create cutting pressure)
- **Hydraulic unit + intensifier** – generates very high pressure (300–400 MPa)
- **Accumulator** – stabilizes pressure and reduces fluctuations
- **Nozzle (orifice)** – converts pressure into high velocity jet
- **CNC system** – controls movement of cutting head
- **Workpiece & catcher tank** – absorbs jet after cutting

Working Principle

- **Water Pressurisation** – A hydraulic pump drives an intensifier piston, which multiplies water pressure up to 300–400 MPa (up to 60,000 PSI)
- **Nozzle Acceleration** – Pressurised water passes through a tungsten carbide or diamond orifice (0.1–0.4 mm diameter), converting pressure to velocity (~900 m/s)
- **Abrasive Mixing (if AWJ)** – Garnet abrasive is drawn into the mixing chamber by venturi effect and entrained in the water jet
- **Material Erosion** – High-velocity jet impacts the ABS surface, eroding material through micro-fracturing and shear – not melting
- **CNC Path Control** – Cutting head moves along programmed tool path at controlled feed rate
- **Spent Jet Collection** – Water and material particles collected in a catcher tank below the cutting bed



Why Waterjet is Suitable for ABS

- **No Heat Generation**

ABS never reaches softening temperature
Eliminates melting and deformation

- **No Mechanical Distortion**

Very low cutting force
No vibration or stress on thin parts

- **Superior Edge Quality**

Clean, smooth edges
No burrs or re-solidified material

- **Improved Sealing Performance**

Flat and consistent edges
Better gasket fitting and reduces leakage

- **High Dimensional Accuracy**

CNC controlled so repeatable results
Tolerance: $\pm 0.1-0.2$ mm



Why Waterjet is Suitable for ABS

Criteria	Waterjet	Laser Cutting	Ultrasonic Cutting
Heat	None	High	Low
ABS suitability	Excellent	Poor	Moderate
Edge quality	Clean	Melted/burnt	Good (thin only)
HAZ	None	High	Low
Thickness	High capability	Limited	<6 mm
Fumes	None	Toxic (HCN)	Minimal



Advantages & Disadvantages

Advantages:

- No heat-affected zone (HAZ)
- High precision (± 0.1 mm)
- Cuts complex shapes in one pass
- No tool wear like routers
- Low cutting force, no deformation
- No toxic emissions
- Reduces rework and finishing operations

Disadvantages:

- High initial cost
- Slower cutting speed compared to laser
- Water and maintenance required
- Parts need drying after cutting



Applications

- Plastic and polymer cutting (ABS, PVC, acrylic)
- Automotive interior panels
- Aerospace composite trimming
- Electronics enclosures (similar to EV charger parts)
- Glass, ceramic, and stone cutting





Problem 3 -Airtight Welding Issue in Industrial Humidifier





Problem Description

The industrial humidifier is fabricated from stainless-steel sheet metal and must remain completely airtight to maintain internal pressure and properly deliver humidified air.

Current Methods Used – conventional MIG and TIG welding

Current Problems

- Inconsistent weld quality between operators and across shifts
- Small gaps and defects forming at sheet-metal joints
- Leakage detected during pressure and leak testing
- Repeated rework and re-welding cycles required
- Distortion of thin stainless steel from the high heat input of the arc



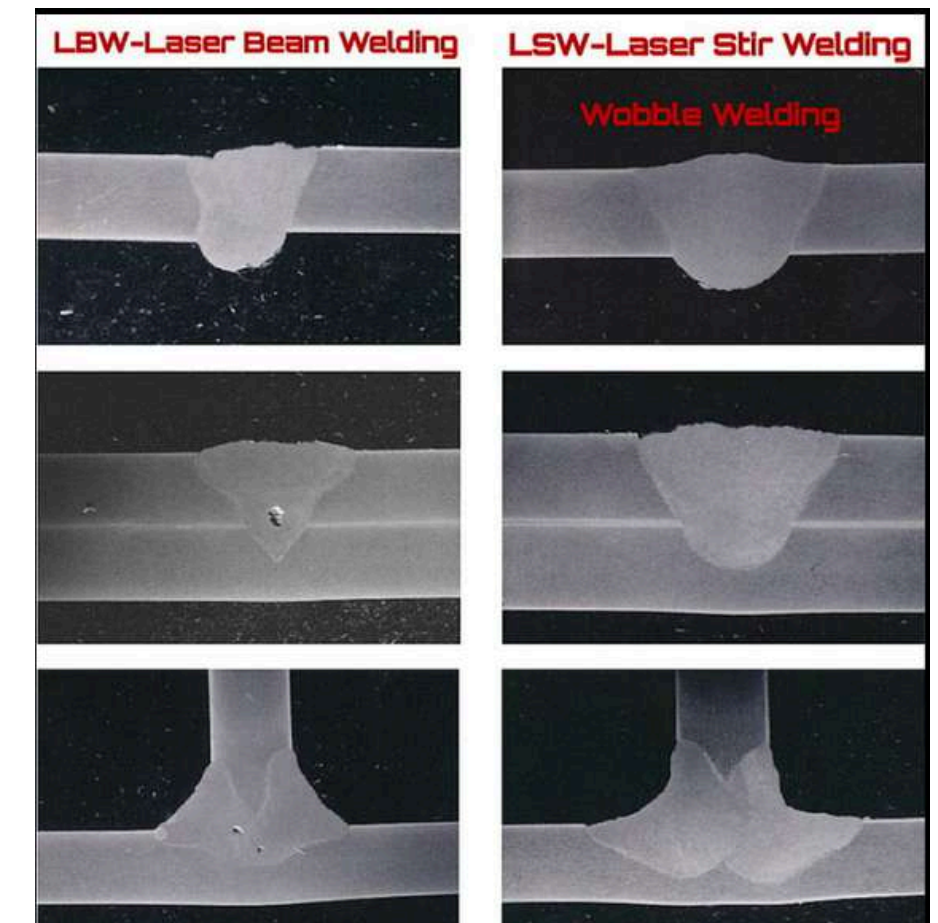
* Limitations of Current Methods

MIG / TIG (arc welding) limitations

- Wide heat-affected zone causes distortion of thin stainless sheet
- Loss of corrosion resistance
- Slow travel speeds reasons for low productivity
- Bead consistency depends heavily on operator skill
- Requires filler wire, flux, and post-weld cleaning

Standard laser welding limitations

- Beam is focused to a fixed, static spot at the focal plane
- Each setup is locked to a specific application no flexibility in bead width
- Material gaps must be limited to ~0.10–0.12 mm, anything larger causes lack of fusion
- Real sheet-metal humidifier joints rarely meet this tight tolerance consistently

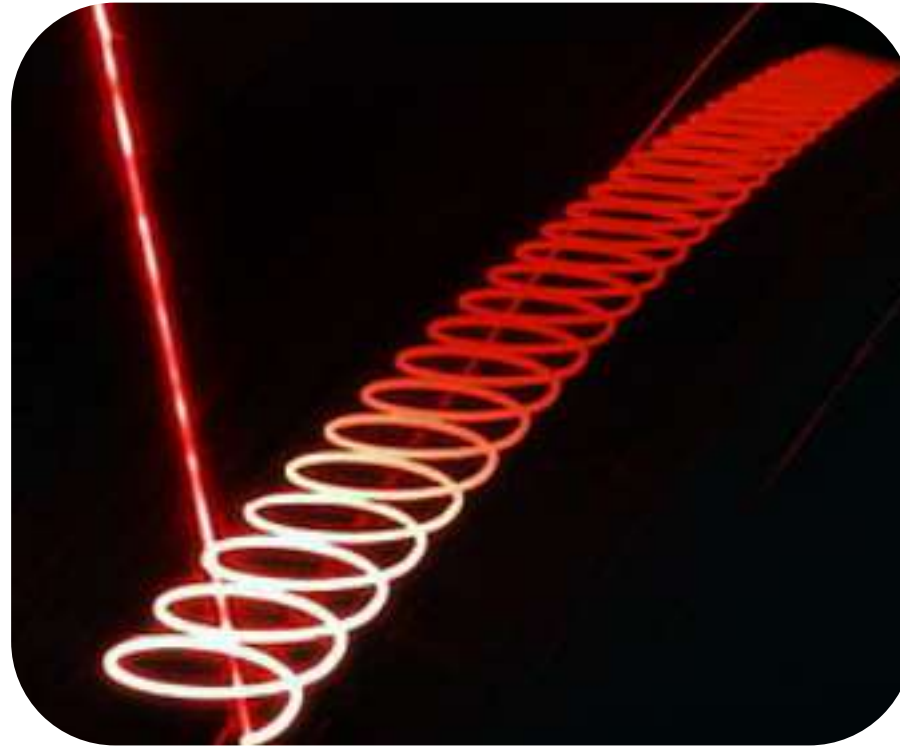


* Why Laser Wobble Seam Welding ?

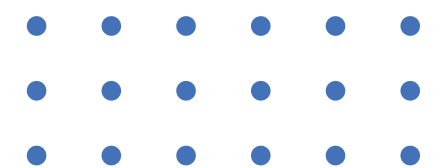
Excellent gap tolerance	bridges fit-up gaps up to 0.5 mm and beyond, with adjustable welding width
Continuous airtight seams	Produces smooth, leak-proof welds with no defects at the start or end of the seam
Low distortion	heat input is typically 200 times lower than other welding processes
High productivity	faster than MIG and TIG welding processes
Strong joints	Creates strong joints without needing extra filler wire, flux, special preparation, or cleaning afterwards.



What Is Laser Wobble Seam Welding?

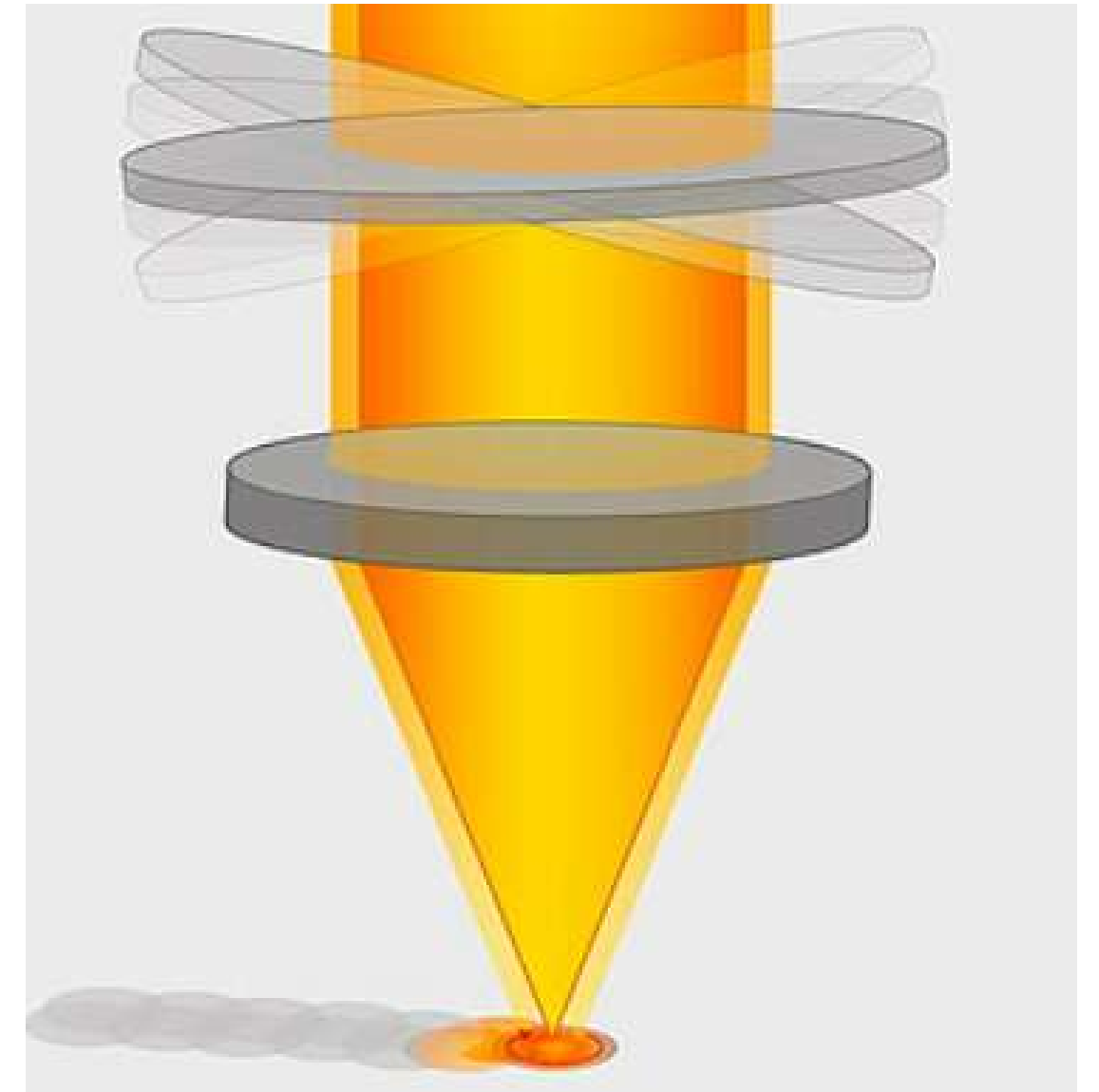


- A laser welding process where the beam is rapidly oscillated (wobbled) while moving along the seam
- Also called laser stir welding — the wobble stirs the molten weld pool
- Wobble heads contain internal scanning mirrors that move the focal spot
- The shape, amplitude, and frequency of oscillation can all be adjusted
- Combines the speed and precision of laser welding with much better gap-bridging tolerance
- Produces a continuous, non-porous joint suitable for airtight applications

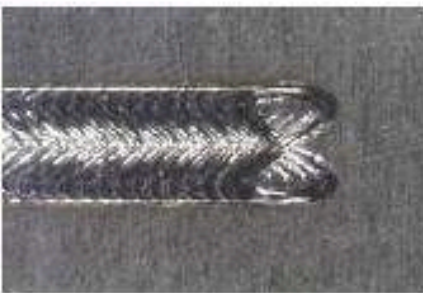
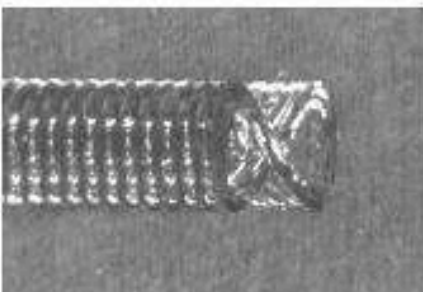


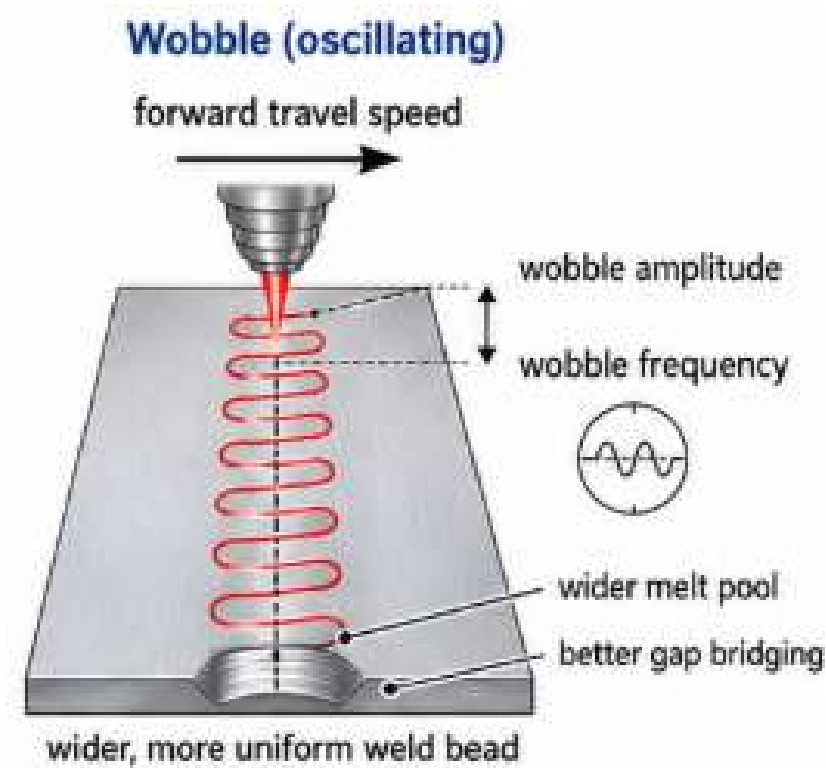
* Working Principle

- Laser photons are absorbed by the metal surface, causing a rapid temperature rise
- Once the surface heats up, absorption increases dramatically a snowball effect that quickly melts the metal and forms a keyhole (vapour cavity)
- The beam is oscillated at high frequency, producing a hydrodynamic stirring action within the keyhole
- The total energy integrated over the oscillation region maintains a large molten pool
- The local intensity of the beam sustains the vapour cavity during stirring
- The rapid motion establishes a self-healing nature of the keyhole



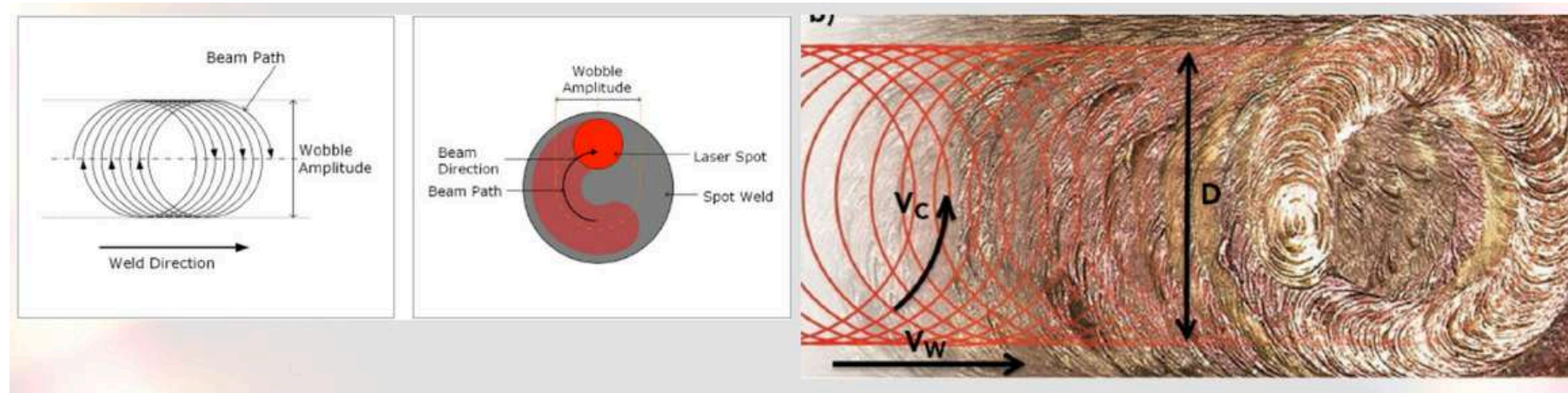
* Wobble Patterns

Wobble Mode	Schematic	Sample Weld
Circle (CW & CCW)		
Linear		
Eight		
Infinity		



* Key Process Parameters

Laser power	higher power gives deeper penetration, too high causes burn-through
Welding speed	affects bead width and heat input
Wobble amplitude	sets the effective seam width and gap-bridging capability
Wobble frequency	controls the stirring effect on the molten pool
Focal position	determines spot size and power density at the joint
Shielding gas	argon prevents oxidation and protects the weld



* Laser Wobble Welding vs Alternatives

Method	Advantages	Limitations
Laser Wobble Welding	Excellent gap tolerance, continuous hermetic seams, very low distortion, faster	High initial cost, requires laser safety enclosure, accurate fixtures needed
Electron Beam Welding	Very high weld quality, deep penetration, minimal distortion	Requires vacuum chamber, very slow cycle time, very high capital cost, limited part size
Friction Stir Welding	Solid-state (no melting), strong joints, low distortion, no filler needed	Needs rigid backing, severe tool wear on stainless steel, not suitable for thin curved sheet, only butt joints





Limitations & Implementation Challenges

- **High initial cost** for the laser source, optics, and motion system
- Class 4 laser hazard requires a **fully enclosed safety** cell with interlocks
- **Accurate fixtures and tight joint** preparation are essential
- Each material and joint must be qualified through **parameter trials**
- **Initial setup and maintenance** require trained personnel
- Long service life and reduced rework offset the high capital cost



Thank You!

